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1 December 1977
SUPERSEDING
MIL-E-1/1081C
7 July 1970

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, KLYSTRON

TYPE 6975

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Reflex, integral cavity, tunable frequency range 8,500 to 9,600 MHz

ABSOLUTE RATINGS:

Parameter:	Ef	Ers	Er	Irs	F	TE	Alt
Unit:	V	Vdc	Vdc	mAdc	MHz	°C	ft
Maximum:	6.9	350	-500	52	9,600	180	60,000
Minimum:	5.7	---	-20	--	8,500	--- (see figure 1)	---

PHYSICAL CHARACTERISTICS: See figure 1

TEST CONDITIONS: See note 1

Parameter:	Ef	Ers	Er	F	TE
Unit:	V	Vdc	Vdc	MHz	°C
	6.3	300	-145 ±5	9,600	140 (max)

GENERAL:

Qualification - Required

(D) Denotes changes

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METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Qualification inspection					
----	Residual frequency modulation	Ef = 5.7 to 6.9 V See Note 15	ΔF	---	0.1	MHz
----	Broadband testing	VSWR = 2.5 (min) See Notes 7 and 15	---	---	---	---
----	Spurious oscillations	See Note 8	---	---	---	---
----	Low pressure	t = 10 seconds See Note 9	ΔF	---	3.0	MHz
4027	Temperature coefficient	TA = +20° to +60°C See Notes 15 and 18	ΔF/ΔT	0	-200	kHz/°C
Ⓓ 1042	Shock	Test condition C, MIL-STD-202, Method 213 See Notes 3 and 15	ΔF	---	2.0	MHz
Ⓓ 1031	Sweep frequency vibration	F = 50 to 1,000 Hz; 10 G; t = 5 minutes See Notes 2, 15 and 17	ΔF	---	1.0	MHz
4250	Power output (2)	F = 8,500 thru 9,600 MHz See Notes 13 and 15	Po	20	---	mW
4280	Electronic tuning range	Power output (2) See Notes 4 and 15	ΔF	30	---	MHz
----	Resonator voltage coefficient	Ers1 = 287.5 Vdc; Ers2 = 312.5 Vdc See Notes 5 and 15	ΔF/ΔV	---	0.15	MHz/V
----	Heater voltage coefficient	Ef = 5.7 to 6.9 V See Notes 6 and 15	ΔF/ΔV	---	4.0	MHz/V
	Quality conformance inspection, part 1	See Note 16				
4229	Total reflector current	TE = 75° ±15°C; See Notes 10, 11 and 14	Ir	---	5.0	μAdc
4214	Cathode emission	Ef = 6.3 to 5.7 V; t = 120; See Notes 11 and 14	ΔIrs/Irs	---	-15	%
1256	Electrode current (resonator)	See Note 15	Irs	---	45	mAdc
4223	Mechanical tuning range	See Note 15 ' Low Freq: High Freq:	F F	---- 9600	8500 -----	MHz MHz
Ⓓ 4250	Power output (1)	F = 8,500 MHz and F = 9,600 See Note 15	Po	20	---	mW
4213	Reflector voltage (1)	See Note 15	Er	-140	-150	Vdc
4213	Reflector voltage (2)	F = 8,500 MHz See Note 15	Er	-85	-105	Vdc
4280	Electronic tuning range	Power output (1)	ΔF	30	---	MHz

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METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 2					
Ⓓ 1031	High frequency vibration	10G; F = 60; t = 120 See note 12	Ir	---	10	μAdc
Ⓓ 4289	Heater current		If	0.40	0.50	A
Ⓓ 4215	Reflector voltage modulation coefficient	See note 15	Coef	1.0	2.0	MHz/V
	Quality conformance inspection, part 3					
----	Life-test provisions	Group D; See note 15	t	500	---	hrs
----	Life-test end points					
4250	Power output (1)	F = 8,500 MHz and F = 9,600 MHz See note 15	Po	15	---	mW

NOTES:

1. All oscillation tests except vibration test shall be made with the tube rigidly connected to a UG-39/U flange, as specified in MIL-F-3922/53, on appropriate RG-52/U waveguide equipment, in accordance with MIL-W-85. The load VSWR for the tube shall be less than 1.1:1. All oscillation tests shall be made under test conditions except when noted.
2. ΔF shall be measured peak-to-peak.
3. The frequency shift after shock in any one position shall not exceed the value specified.
4. Superimposing a 60 Hz ac voltage sweep of 60 volts, peak-to-peak, on the direct-reflector voltage, the mode curve shall be observed on an oscilloscope whose amplifier has a minimum bandpass of 0.1 MHz. The power output shall have no discontinuities due to hysteresis or spurious mode effects between half-power points for both directions of reflector voltage change.
5. The resonator voltage coefficient shall be defined as the change in frequency as measured at the maximum power point of the reflector voltage mode divided by the corresponding static change in resonator voltage.
6. The heater voltage coefficient shall be defined as the change in frequency as measured at the maximum power point of the reflector voltage mode divided by the corresponding change in heater voltage.
7. Superimposing sufficient 60 Hz ac voltage on the direct-reflector voltage to suppress oscillations on the ends of the sweep, the mode curve shall be observed on an oscilloscope. With a standing-wave introducer inserted, there shall be no discontinuities at the maximum power points for any phase of the standing wave. If the center of the mode is distorted a wavemeter pip should not disappear for any interval of the mode center as the wavemeter is tuned to check for a frequency discontinuity.
8. The frequency of any existing spurious modes of oscillation shall not be within the frequency range of 8,400 to 9,700 MHz for any position of the tuning screw. The power output of any existing spurious modes of oscillation shall not exceed 15 milliwatts.

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NOTES: (Cont'd)

9. The frequency shall be stabilized at a pressure of 70 mmHg. The pressure shall be increased to 730 to 760 mmHg in less than the time specified. The resulting frequency change shall not exceed the limit specified.

10. After 2 minutes with all voltages applied, total reflector current shall not exceed the specified limits.

11. The tube shall not be oscillating during the test.

12. The tube shall be vibrated at 10 G at a frequency of 60 Hz for 2 minutes in three mutually perpendicular planes. The reflector current bursts shall be recorded with an instrument whose frequency response is from dc to 100 Hz, with not more than 3 dB fall-off at 100 Hz. There shall be no reflector current bursts greater than the limits specified when measured across a 10,000 ohm resistor.

13. The tube under test will be tested over the specified frequency range at 100 MHz intervals.

14. This test to be performed at the conclusion of the holding period.

15. Lr (mode 6-3/4) adjusted for maximum power output.

16. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0 percent, inspection level II.

D 17. The tube shall be vibrated in each of three mutually perpendicular planes over the frequency range of 50 to 1,000 Hz at logarithmic sweep. The frequency deviation resulting from the vibration, shall not exceed the limits specified. The tubes shall meet all quality conformance inspection, part 1 tests following vibration.

18. For the measurement of temperature coefficient, the tube shall be adjusted in free air to the frequency indicated under test conditions. The reflector voltage shall be adjusted for maximum power for the mode specified. The tube shall be completely enclosed with a metallic cover with an approximate clearance of 1/2 inch (12.7 mm) between the tube and the cover. The walls of the container shall be heated so that an ambient temperature of 55° to 60°C is obtained within. The frequency of oscillation is determined and the walls then cooled at a rate such that the fall of ambient temperature within shall not exceed 2°C per minute. When the ambient temperature has been reduced to 25°C or lower, the frequency shall then be recorded. This procedure shall be repeated three times with no readjustment of the frequency on the second two runs. The temperature coefficient shall be computed in frequency units per degree Celsius from the average obtained on the last two runs.

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Review activities:

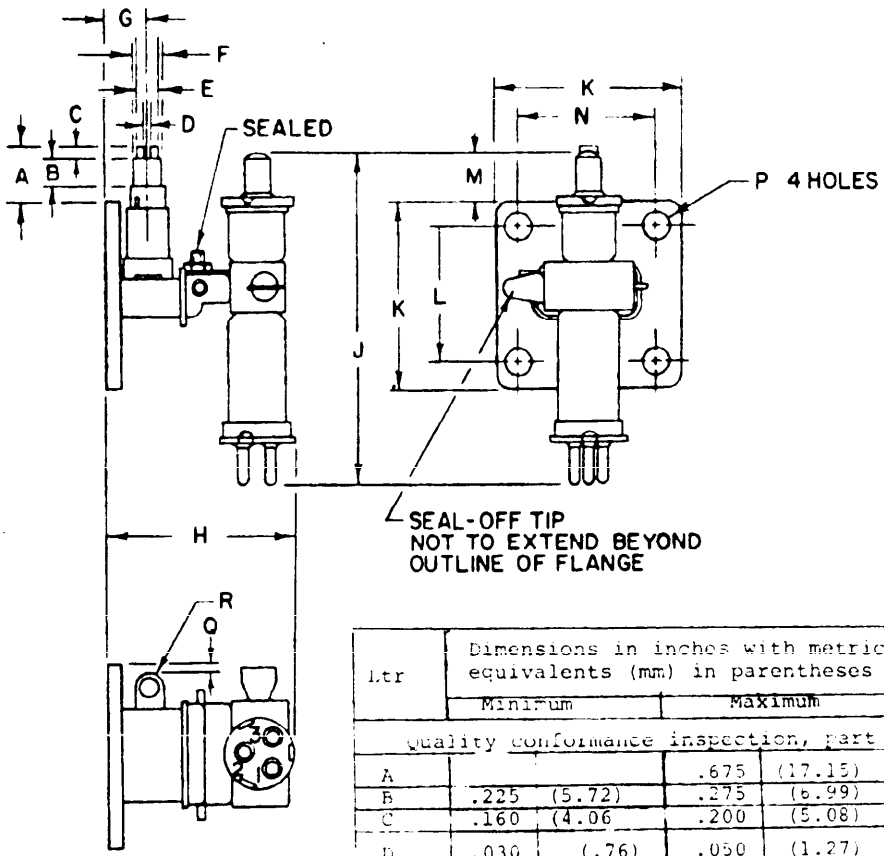
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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses			
	Minimum		Maximum	
Quality conformance inspection, part 1				
A			.675	(17.15)
B	.225	(5.72)	.275	(6.99)
C	.160	(4.06	.200	(5.08)
D	.030	(.76)	.050	(1.27)
E	.177	(4.50)	.197	(5.00)
F	.245	(6.22)	.250	(6.35)
G	.375	(9.53)	.425	(10.80)
H			1.575	(40.01)
J			3.000	(76.20)
K	1.615	(41.02)	1.635	(41.53)
L	1.276	(32.41)	1.284	(32.61)
M			.450	(11.43)
N	1.216	(30.89)	1.224	(31.09)
P	.214	(5.44)	.223	(5.66)
Q	.050	(1.27)	.100	(2.54)
R	.135	(3.43)	.150	(3.81)

NOTES:

a. Unless otherwise specified, the AQL for all tests listed under Quality Conformance Inspection, part 1, shall be 1.0 percent, inspection level II.
b. Cathode: Unipotential
c. Cap: C1-4.
d. Base: A3-108.
e. Flange mates with UG-39/U (MIL-F-3922/53).
f. Envelope temperature (TE) to be measured on the ground lug.

FIGURE 1. Outline drawing of electron tube type 6975.

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